

QUALIFYING EXAMINATION, Part 1

Solutions

Problem 1: Classical Mechanics I

(a) $\vec{R} = \frac{1}{M} \sum_i m_i \vec{r}_i$ where $M = \sum_i m_i$.

(b)

$$\begin{aligned} \vec{L} &= \sum_i m_i \vec{r}_i \times \dot{\vec{r}}_i = \sum_i (\vec{r}_i + \vec{R}) \times m_i (\dot{\vec{r}}_i + \dot{\vec{R}}) \\ &= M \vec{R} \times \dot{\vec{R}} + \sum_i m_i (\vec{r}_i \times \dot{\vec{r}}_i) + \sum_i (m_i \vec{r}_i) \times \dot{\vec{R}} + \vec{R} \times \sum_i m_i \dot{\vec{r}}_i \\ &= \vec{L}_{CM} + \vec{L}_{rel} + 0 + 0, \end{aligned}$$

where we used $\sum_i m_i \vec{r}_i = 0$ in the CM frame.

(c)

$$\vec{v} = \hat{x}v, \quad \vec{V} = \hat{x}V, \quad \vec{\omega} = \hat{z}\omega$$

We choose the fixed origin O to be the lower end of the rod (before it moves). The angular momentum $\vec{L}$ around O is conserved (since there is no torque). The mass m does not contribute to $\vec{L}$, and by conservation of $\vec{L}$, we find

$$0 = -M \frac{l}{2} V + \frac{Ml^2}{12} \omega,$$

where on the r.h.s. we used the equation in (b) to calculate L_z of the rod with respect to O and $\frac{Ml^2}{12}$ is the moment of inertia of the rod around O . We find

$$\omega = \frac{6V}{l}.$$

By the conservation of linear momentum along $\hat{x}$, we have

$$mv = mv' + MV.$$

In addition, the velocity v' is also the velocity the lower end of the rod right after the collision, and thus

$$v' = V + \omega \frac{l}{2}.$$

Substituting in the equation linear momentum conservation and using the above equation for ω , we find

$$V = \frac{mv}{(4m + M)},$$

and

$$v' = 4V = \frac{v}{1 + (M/4m)} .$$

(d) Kinetic energy will be conserved and the mass m will have an independent position.

Problem 2: Classical Mechanics II

(a) The transformation equations (expressing the cartesian coordinates in terms of the generalized coordinates) are

$$x = \rho \cos \varphi, \quad y = \rho \sin \varphi, \quad z = \frac{1}{4a^3} \rho^4.$$

The cartesian velocities are given by

$$\dot{x} = \dot{\rho} \cos \varphi - \rho \dot{\varphi} \sin \varphi, \quad \dot{y} = \dot{\rho} \sin \varphi + \rho \dot{\varphi} \cos \varphi, \quad \dot{z} = \left(\frac{\rho}{a}\right)^3 \dot{\rho}.$$

Then the kinetic energy is

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \frac{1}{2} m \left[\dot{\rho}^2 \left(1 + \left(\frac{\rho}{a}\right)^6 \right) + \rho^2 \dot{\varphi}^2 \right].$$

The bead's potential energy in the gravitational field is

$$V = mgz = mg\rho^4/(4a^3).$$

The Lagrangian is therefore

$$L = T - V = \frac{1}{2} m \left[\dot{\rho}^2 \left(1 + \left(\frac{\rho}{a}\right)^6 \right) + \rho^2 \dot{\varphi}^2 \right] - \frac{1}{4a^3} mg\rho^4.$$

(b) Recall that for a generic generalized coordinate q , the Euler-Lagrange equation of motion is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0.$$

Applying this result to (ρ, φ) gives,

$$\begin{aligned} \frac{d}{dt} \left[m\dot{\rho} \left(1 + \left(\frac{\rho}{a}\right)^6 \right) \right] &= 3m \left(\frac{\rho^5}{a^6} \right) \dot{\rho}^2 + m\rho \dot{\varphi}^2 - mg \left(\frac{\rho}{a} \right)^3, \\ \frac{d}{dt} (m\rho^2 \dot{\varphi}) &= 0, \end{aligned}$$

as the equations of motion for ρ and φ , respectively.

(c) The system is invariant under rotations around the z -axis $\varphi \rightarrow \varphi + \Delta\varphi$, as well as under time translations $t \rightarrow t + \Delta t$, where $\Delta\varphi$ and Δt are continuous real parameters. Therefore, according to Noether's theorem, the generators of these continuous symmetries are constants of the motion. The generator of rotations around the z -axis is the angular momentum component

$$L_z \equiv \frac{\partial L}{\partial \dot{\varphi}} = m\rho^2 \dot{\varphi},$$

while the generator of translations in time is the Hamiltonian

$$H \equiv \left(\frac{\partial L}{\partial \dot{\rho}} \right) \dot{\rho} + \left(\frac{\partial L}{\partial \dot{\varphi}} \right) \dot{\varphi} - L = \frac{1}{2} m \left[\dot{\rho}^2 \left(1 + \left(\frac{\rho}{a} \right)^6 \right) + \rho^2 \dot{\varphi}^2 \right] + \frac{1}{4a^3} m g \rho^4 = T + V.$$

Indeed, the conservation of L_z is just the second equation of motion in (b) (φ is a cyclic coordinate). Note that the velocity in the φ direction is $\rho\dot{\varphi}$, so L_z is just the angular momentum about the z -axis.

The fact that $H = T + V$ identifies H as the conserved total energy of the bead. The conservation of H also follows from $dH/dt = -\partial L/\partial t = 0$.

(d) For a circular orbit about the z -axis with $\rho = \rho_c$, $\dot{\rho} = 0$, the first equation of motion in (b) implies that $\dot{\varphi}^2 = g\rho_c^2/a^3$ is a constant. Since $\dot{\varphi}$ is the angular velocity of the circular orbit, the period P is given by

$$P = \frac{2\pi}{|\dot{\varphi}|} = \frac{2\pi a}{\rho_c} \sqrt{\frac{a}{g}}.$$

(e) Eliminating $\dot{\varphi}$ in favor of L_z , we see that the total energy of the system can be expressed as

$$H = \frac{1}{2} m \dot{\rho}^2 \left(1 + \left(\frac{\rho}{a} \right)^6 \right) + V_{eff}(\rho),$$

with $V_{eff}(\rho) = \frac{L_z^2}{2m\rho^2} + \frac{1}{4a^3} m g \rho^4$ an effective potential that describes the radial motion of the system. The radial motion is influenced by two competing effects: an angular momentum barrier $\propto 1/\rho^2$ (for $L_z \neq 0$) that repels the particle away from the origin and a gravitational potential $\propto \rho^4$ which prevents the particle's radial coordinate from increasing to $\rho \rightarrow \infty$ without bound. As a result, we expect the system to exhibit *stable bound orbits* that wind around the origin $\rho = 0$.

The stability of the circular orbit with respect to small perturbations can also be inferred by eliminating $\dot{\varphi}$ in favor of L_z in the equation of motion for ρ , and then linearizing this equation around the circular orbit of (d). Defining $\rho = \rho_c + \delta\rho$, we find for $\delta\rho \ll \rho_c$ that the LHS of the first equation in (b) can be approximated as

$$\text{LHS} \approx m \left(1 + \left(\frac{\rho_c}{a} \right)^6 \right) \frac{d^2}{dt^2} \delta\rho,$$

and

$$\begin{aligned} \text{RHS} &= 3m \left(\frac{\rho_c^5}{a^6} \right) \left(1 + \frac{\delta\rho}{\rho_c} \right)^5 (\delta\dot{\rho})^2 + \frac{L_z^2}{m\rho_c^3} \left(1 + \frac{\delta\rho}{\rho_c} \right)^{-3} - mg \left(\frac{\rho_c}{a} \right)^3 \left(1 + \frac{\delta\rho}{\rho_c} \right)^3 \\ &\approx -\frac{6mg\rho_c^2}{a^3} \delta\rho + \mathcal{O}(\delta\rho^2), \end{aligned}$$

where the result $L_z^2/m = mg\rho_c^6/a^3$ from (d) has been used. Setting LHS=RHS, we see that the radial perturbation satisfies the equation of a simple harmonic oscillator,

$\frac{d^2}{dt^2}\delta\rho = -\omega_\rho^2\delta\rho$, where the frequency of radial oscillations is given by

$$\omega_\rho^2 = 6g\rho_c^{-1} \left[\left(\frac{\rho_c}{a} \right)^3 + \left(\frac{a}{\rho_c} \right)^3 \right]^{-1}.$$